Thakur Trusts (Regd.)
THAKUR COLLEGE OF
ENGINEERING & TECHNOLOGY
Autonomous College Affiliated to University of Mumbai
Approved by All India Council for Technical Education (AICTE) and Government of Maharashita(GOM)
Conferred Autonomous Status by University Grants Commission (UGC) for 10 years w.e.f. A.Y. 2019-20
(among the First in the country to be conferred Autonomous Status) and awarded 'A+' Grade by NAAC (3rd Cycle)
(Institute Ranked 58th in India by NIRF, Ministry of Education, Government of India)
Estd. 2001

IN-SEMESTER EXAMINATION-I

F.E./F.T. (Semester-I)

SUBJECT - Physics

Branch: COMP, CIVIL, E&CS, CSE, AI&DS, IoT **Date: 24/10/2024**
Div.: ALL **Timing: 10:00 AM to 11:00 AM**
Duration: 60 Minutes **Maximum Marks: 20**

Instructions –

1. All questions are compulsory.
2. Assume suitable data wherever necessary and state the assumptions made.
3. Diagrams / sketches should be given wherever necessary.
4. Use of logarithmic table, drawing instruments and non-programmable calculators is permitted.
5. Figures to the right indicate full marks.

Q.1		Answer any 5 of the following questions	Marks	Course Outcomes	Learning Levels
	a.	Calculate electron & hole concentrationn in intrinsic silicon at room temperature if it's electrical conductivity is 4*10^4 mho/m.Given that mobility of electron = 0.14 m2/V-sec & mobility of holes = 0.04m2/V-sec	2	1	A
	b.	The mobility of holes is 0.025m2/V sec. What would be the resistivity of p type Silicon if the Hall coefficient of the sample is 2.25*10^5 m3/C	2	1	A
	c.	Write down the advantages of PIN photodiode over pn photodiode.	2	1	R
	d.	Write any 4 Characteristics of matter waves.	2	2	U
	e.	What is the de Broglie wavelength of an electron of kinetic energy 500 eV?	2	2	A
	f.	Explain the significance of calibration in the context of sensor accuracy.	2	3	R
	g.	Provide two examples of light sensors and discuss one application for each.	2	3	A
Q.2	a.	What is the wave function? Write the physical significance of the quantum mechanical wave function. What are the properties of well-behaved wave function?	5	2	U
		OR			
	b.	Calculate the de-Broglie wavelength associated with an alpha particle accelerated by potential difference of 30kV. Mass of alpha particle is 6.68×10^{-27} kg	5	2	U
Q.3	a.	Discuss the principle, construction, and working of a light-dependent resistor (LDR). Include a diagram and elaborate on four applications where LDRs are commonly used.	5	3	R
		OR			
	b.	What is the difference between Active & Passive sensors? Explain active & passive sensors with examples.	5	3	U